

A Model of Optimal QE and QT with Liquidity Regulations

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December 4, 2023

Motivation

- In May 2022, the Fed initiated a new round of quantitative tightening.
 - ▶ Current balance sheet size: \$8 tn. Mostly treasuries (\$4.2 tn) and MBS (\$2.5 tn).
 - ▶ Monthly reduction cap: \$90 bn for treasuries and \$35 bn for agency MBS.
 - ▶ Will slow down at *"the level consistent with **ample reserves**"*.
- COVID QE/QT happens in a different financial system than GFC QE.
 - ▶ Higher demand for reserves with new regulations (Lopez-Salido and Vissing-Jorgensen [2023]).
 - ▶ Concerns about liquidity stress associated with large balance sheet and QT (Acharya et al. [2023]).
- Research question: what is the optimal asset-purchase and **exit** policy?
 - ▶ What is the optimal unwinding speed: potentially state contingent. (Karadi and Nakov [2021])
 - ▶ Where to stop/slow down: what does the level consistent with ample reserves mean?

This paper

- A DSGE model of QE/QT with financial intermediaries (banks)
 - ▶ Segmented bond market due to bank's inelastic bond demand.
 - ▶ Intermediary wealth matters for leverage and productivity. (Brunnermeier and Sannikov [2014])
 - ▶ QE/QT affects recovery of bank net worth. (Gertler et al. [2012])
- Exploring the role of new liquidity regulations
 - ▶ **Liquidity Stress Testing** (LST): reserves $\downarrow \Rightarrow$ intermediation efficiency $\downarrow$. (d'Avernas and Vandeweyer [2020])
 - ▶ **Leverage Coverage Ratio** (LCR): outflow risk must be buffered by liquid assets (HQLAs).
 - ▶ Reserves $\Rightarrow$ bond prices & HQLAs holding $\Rightarrow$ leverage, investment.
- Ramsey problem: the central bank chooses its balance sheet size.
 - ▶ Trade-off between output and balance sheet cost.
 - ▶ Optimal QT slows down when LCR binds.

Environment

- Discrete time, quarterly, infinite horizon, representative agents.
- **Firm** produces goods by $Y = AzK^\alpha L^{1-\alpha}$
 - ▶ Productivity z follows a log $AR(1)$ process.
 - ▶ Capital K is an CES aggregator of bank's and household's capital
- **Bank** with financial technology:
 - ▶ Assets: invest in capital K^b , hold reserves R , and government bonds qB^b (perpetual bond).
 - ▶ Liabilities: deposit D^b and net worth N^b .
- **Household** with period utility $u(C^h, D^h) = \log(C^h) + \gamma \log(D^h)$
 - ▶ Assets: government bond qB^h , bank deposit D^h , and capital K^h (less productive).
 - ▶ Unit labor endowment for each period.
- **Central bank** sells/buys bonds with reserves, transferring gain/loss to the government.
- **Government** finances a constant spending and coupon payments with CB remittance & lump sum taxes.

Agents' Balance Sheets

	Banks		Households		Central bank		Government	
HQLAs	Reserve	Deposit	Bond	Tax	Bond	Reserve	Tax	Bond
	Bond		Deposit			TGA	TGA	
	Capital	Net worth	Capital	Net worth				

Agents' problem

- Bank maximizes utility $\sum_{t=0}^{\infty} \beta_b^t \log(C_t^b)$ subject to budget constraint

$$N_t^b = R_{t-1}^b (1 + r_t) + (q_t + 1)B_{t-1}^b + K_{t-1}^b (1 + r_t^k - \delta) - D_{t-1}^b (1 + r_t^d) \\ - \underbrace{\eta \exp \left(\psi \frac{q_{t-1} B_{t-1}^b}{R_{t-1}^b} \right)}_{\text{LST costs}} - \underbrace{\kappa \left(\frac{D_{t-1}^b}{N_{t-1}^b} \right)^2}_{\text{Balance sheet costs}} - C_t^b - \epsilon_t^N$$
$$N_t^b = R_t^b + q_t B_t^b + K_t^b - D_t^b$$

- ▶ Reserves decrease $\Rightarrow$ intermediation efficiency declines. (**LST**) [▶ regulation details](#)
- ▶ Deposit must be backed by high quality liquid assets (HQLAs). (**LCR**) [▶ regulation details](#)

$$\varphi D^b \leq L := (R + \omega q B^b)$$

- ▶ Bond liquidity ω is subject to an $AR(1)$ liquidity shock.

Agents' problem

- Household maximizes utility subject to

$$K_t^h + D_t + q_t B_t^h = K_{t-1}^h (1 + r_t^{k,h}) + D_{t-1}^h (1 + r_t^d) + (q_t + 1) B_{t-1} - \frac{\chi}{2} (B_t^h - q_{ss} \bar{B}_h)^2 + W_t - T_t$$

- Government budget constraint

$$T_t + \underbrace{\left(\frac{q_t}{q_{t-1}} - 1 \right) \bar{R}_{t-1}}_{\text{capital gain/loss on bonds}} - \underbrace{r_t \bar{R}_{t-1}}_{\text{interest payed on reserves}} = \bar{B}_{t-1} + G_t, \quad G_t = G_{ss}$$

- The central bank implements QE/QT by buying/selling bonds. $\bar{B}_t = B_t^b + B_t^h$

$$q_t (\bar{B}_t - \bar{B}_{t-1}) + \bar{R}_t - \bar{R}_{t-1} = 0$$

- Ramsey problem:

$$\mathcal{L} = \min_{\{\bar{R}_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta_h^t \left((Y_t - Y_{ss})^2 + \nu (\bar{R}_t - \bar{R}_{ss})^2 \right)$$

Mechanism

- Bank bond pricing: LST cost vs convenience yield

$$\mathbb{E}_t \left[\Lambda_{t+1}^b \frac{q_{t+1} + 1}{q_t} \right] = 1 + \eta \exp \left(\psi \frac{q B^b}{R_t^b} \right) - \omega \frac{\lambda}{\varphi}$$

- From households' perspective

$$\mathbb{E}_t \left[\Lambda_{t+1}^h \frac{q_{t+1} + 1}{q_t} \right] = 1 + \chi \left(q_t B_t^h - q_{ss} \bar{B}^h \right)$$

- QT's effect when LCR does not bind

- ▶ LST $\Rightarrow$ banks' inelastic bond demand $\Rightarrow$ Bond price falls when supply increases.
- ▶ $q \downarrow$ translate one-for-one to wealth $\downarrow \Rightarrow$ leverage cost $\uparrow \Rightarrow$ leverage goes down.

- QT's effect when LCR binds

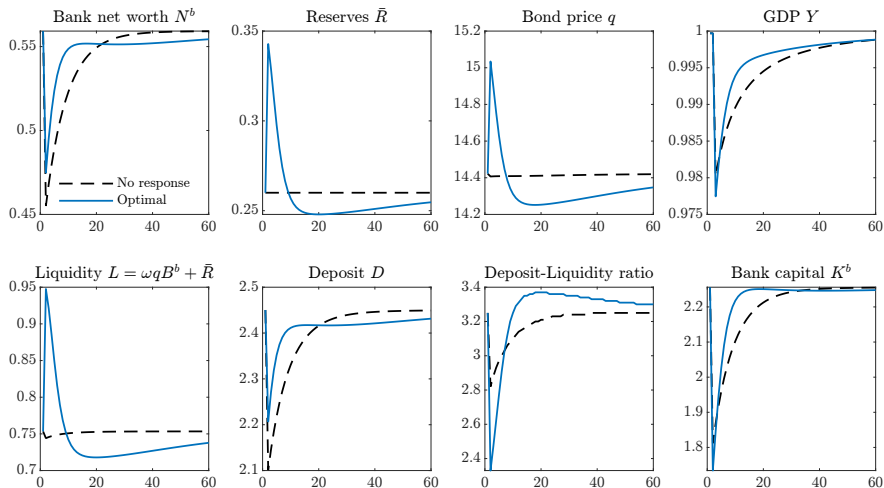
- ▶ LST $\Rightarrow$ banks' bond holdings (HQLAS) $\downarrow$ as reserves decrease $\Rightarrow$ (LCR) leverage goes down.
- ▶ Change in q is ambiguous.

Parameter Values

Param'r	Value	Description	Target/moment	Model
A. Households				
β_h	0.99	HH's discount factor	Annual rate of 4%	
$q_{ss}\bar{B}_h$	2	HH's bond target	Share of bond held by HH	77%
χ	0.1	Cost of holding bond	Bond demand elasticity	0.36
γ	0.02	Liquidity service	Bank's quarterly profit margin	2.5%
B. Bankers				
β_b	0.97	Bankers' discount factor	Bank net worth to GDP ratio 10.2%	13%
κ	0.01	Leverage ratio cost	S.S. bank leverage ratio 8.0	4.3
(η, ψ)	$(10^{-4}, 2.5)$	LST cost function params	Security holding to reserve ratio 1.7	2.0
φ	1/2.5	LCR constraint	Deposit to HQLAs ratio 2.3	2.5
C. Firm and government				
(A, α)	$(0.53, 0.33)$	Ave. productivity; capital share	Unit quarterly GDP; labor share 67%	
(ϵ, θ)	$(3, 0.9)$	CES capital aggregator params		
δ	0.025	Depreciation rate		
$q_{ss}\bar{B}_{ss}$	3	S.S. private bond holding	Private sector bond to GDP ratio 78%	75%
$\bar{R}_{ss}$	0.26	S.S. reserve level	Reserves to GDP ratio 6.5%	6.5%
ν	0.01	Balance sheet cost		

Results: net worth shock (slack LCR)

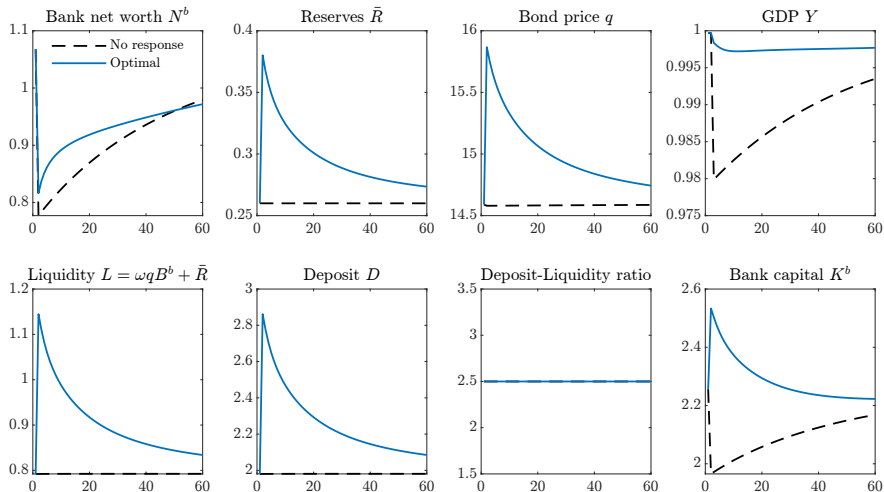
- QE inflates bond price $\rightarrow$ Higher net wealth $\rightarrow$ lower leverage cost $\rightarrow$ faster recovery.



Results: net worth shock (binding LCR)

- The liquidity (LCR) margin instead of the leverage ratio margin. Optimal QT is slower.

►► liquidity shock



Conclusion

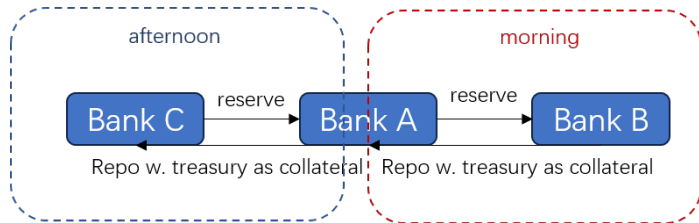
- Under the ample-reserves regime with current regulations:
 - ▶ QE can mitigate financial and productivity shocks by restoring bank net worth.
 - ▶ More powerful when liquidity is the cause of intermediation failure (binding LCR).
 - ▶ QT only has to slow down when liquidity is scarce.
- Next steps:
 - ▶ Micro found the policy objective.
 - ▶ Coordination with interest rate policy with NK production side.
 - ▶ Break the multiple roles of "treasury bond" (money and treasury market).

Reference

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Appendix: Liquidity Stress Testing [» back](#)

- LST: **short-term** outflow risk must be buffered by reserves.
- Bank does not want to go below the LST redline of reserve holdings at anytime.
- Intra-day outflow can happen due to the difference in timing of large repo transactions/payments.
- Higher repo market size → Higher intra-day outflow → more likely to break the redline.
- Intermediation costs increase when there are fewer reserves.
- In the model: costs of holding treasury bonds increases when there are fewer reserves.

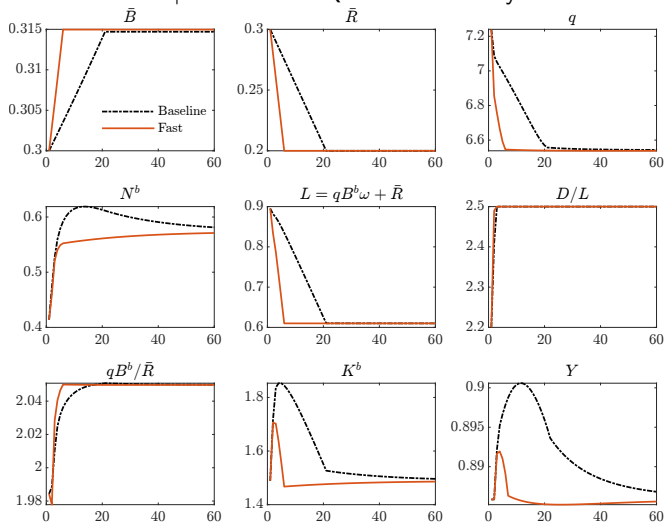


Appendix: Liquidity Coverage Ratio [▶▶ back](#)

- Basel III: banks must hold high-quality liquid assets (HQLAs) with a minimum of **100%** of the net outflow over a **30-day** stress period.
- HQLAs include: Reserves, Treasuries (no haircut), and other assets like government-backed MBS (haircut up to 40%).
- The regulation was proposed in 2010, but the 100% minimum was not enforced until 2019.
- A more realistic assumption than reserve requirement.
- In the model: $HQLA \leq \varphi D$
- Interaction with LST: reserve $\downarrow \rightarrow$ repo $\downarrow$ (LST) $\rightarrow$ HQLAs $\downarrow \rightarrow D \downarrow$ (LCR).

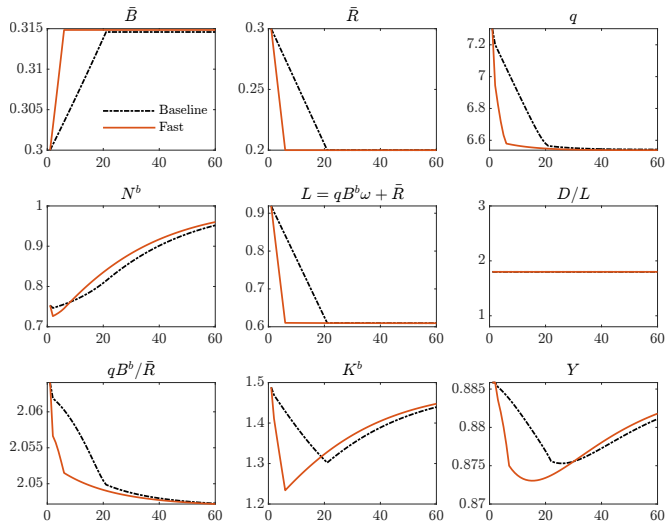
Results: unit root QT shock (slack LCR)

- QT is slightly expansionary. The slower, the more expansionary.
 - ▶ Lower reserves $\rightarrow$ LST cost $\uparrow \rightarrow$ Lower HQLAs $\rightarrow$ Directly substitute capital for bonds.



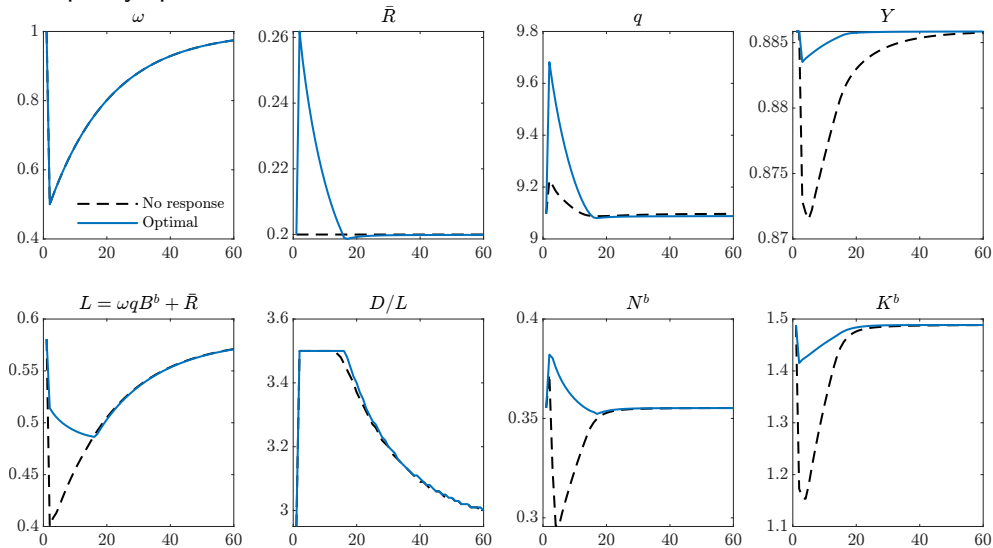
Results: unit root QT shock (binding LCR)

- QT is contractionary. The faster, the more contractionary



Results: liquidity shock (slack LCR) [▶▶ back](#)

- Optimal policy: provide reserve until the constraint does not bind.



Results: productivity shock (slack LCR)

- channel here.

